

Which secrets are still possible?

Name: _____

Date: _____

Two colors: R and B. Repeats allowed. A test earns one point for each color in the correct position. You hear **only the total**, not which positions match.

1. Learn the checker job.

Use two counters. First leave the secret visible. Try some tests. Check one place at a time, then say only the total. Now hide the secret and play.

2. Make all four secrets.

Build these four rows. Use different shapes as well as colors if you like.

RR

RB

BR

BB

3. One answer rules some out.

The test RR scores **1**. Cover every secret that cannot fit. How many remain?

Draw or point to the secrets that remain.

4. Choose a useful next test.

What test could tell the remaining secrets apart? Try it with both possible secrets.

My test; what its different answers would tell me:

Build first. Explain by pointing, talking, or drawing.

Make a secret-finding tree

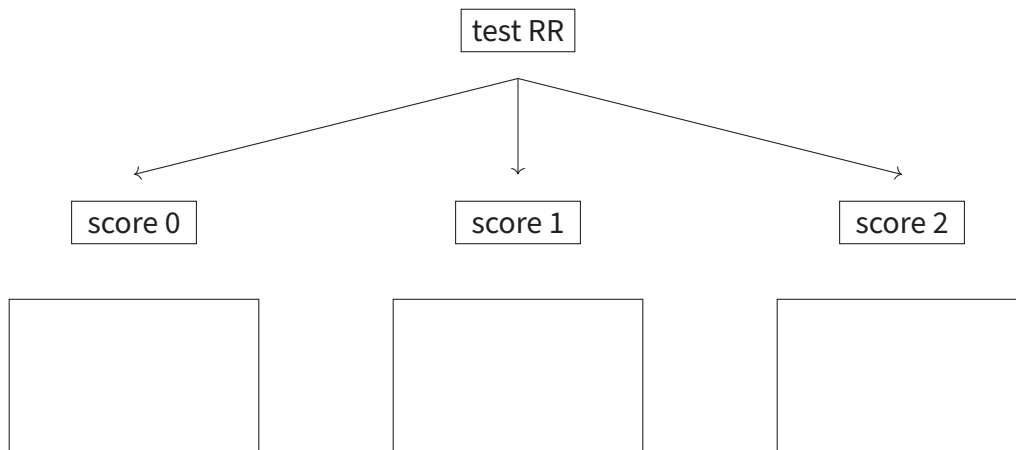
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5. Begin with RR.

An adult helps record the three things you might hear. Put possible secrets below each answer.



6. Finish every branch.

If two secrets remain, choose a test that separates them. If only one remains, you can name it without another test.

Could one test always tell the secret? Show a troublesome answer.

Play again: Exchange maker and finder jobs. Can you follow your tree for every secret? An adult can do the writing; you supply the reasons.

Build first. Explain by pointing, talking, or drawing.